BLACK PEN Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

LESSON 031 · D-type flip-flops

Name: _____ Class: _____ Date: _____

Work through the pages in order. Answer in BLACK PEN first. Check in GREEN PEN only after the Exit Ticket.

Do Now · Recall previous learning

Q1. A half adder receives A = 1 and B = 0. Calculate Sum and Carry. Explain which logic operation produces each output.

Q2. Apply De Morgan’s law to simplify ¬(A ∨ B).

Q3. Insert 40, 20, 60, 30 into an empty binary search tree. State the parent of 30 and whether 30 is its left or right child.

Q4. A = $0.1100_2 \times 2^3$ and B = $0.0100_2 \times 2^3$. Calculate A − B, showing the mantissa calculation and decimal result.

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I Do · Read new knowledge

Key vocabulary

Recall the term — D-type flip-flop

Recall the term — D input

Recall the term — clock

Recall the term — trigger

Recall the term — Q output

Recall the term — state

Recall the term — edge

New learning

Why do we need flip-flops? — Logic gates calculate; a flip-flop remembers. Registers must hold values even when other inputs change.

Meet the D-type flip-flop — D is the data input. Q is the stored output. CLK determines when Q can update.

Supermarket door analogy — D can change before the sample. Q only stores the new value at CLK ↑.

Digital watch: divide by two — NOT Q fed into D causes Q to toggle at each rising edge.

The rising edge matters — In this lesson we use a rising-edge-triggered D flip-flop: Q copies D only when CLK changes from 0 to 1.

Between clock edges, Q holds — Changing D alone does not change Q. Without a rising edge, the flip-flop keeps its previous bit.

Locate the rising edges — Q initially stores 0. Predict Q after the first rising edge.

Trace rising edge 1 — At ↑1, D = 1. Q becomes 1.

Worked example

Model when Q changes — and when it holds

1. Identify which edges count — Point to CLK: ↑ is LOW → HIGH and samples D. ↓ is HIGH → LOW and does not update Q. Find all three ↑ edges before reading D.

2. ↑1: Q changes from 1 to 0 — CLK rises. Read D = 0. Q was 1, so copying 0 causes a visible change. On the following falling edge Q holds 0, even though CLK moves.

3. D changes, but Q stays 0 — Look where D rises between ↑1 and ↑2. The clock has not risen again, so Q does not follow D. The falling clock edge also leaves Q unchanged.

4. ↑2: Q changes from 0 to 1 — CLK rises again. D is now 1, and Q was 0. Sampling D makes Q change to 1. Q holds at 1 after CLK falls.

5. ↑3: Q stays 1 — despite a rising edge — CLK rises and samples D = 1. But Q already stores 1, so the output does not visibly change. The rule is: at ↑ copy D; otherwise hold Q.

We Do · Practise with guidance

Trace four rising clock edges

Q initially stores 1. Predict its value at each rising edge (↑).

Predict edge 1

Find D at ↑1. What should Q store next?

D = 0 → Q = 0

Q changes from 1 to 0. Complete row 1.

D = 1 → Q = 1

Q updates from D at ↑2. Complete row 2.

Only a rising edge causes Q to sample D. Complete row 3.

Q becomes 1. Record row 4 and review the complete waveform.

Rising edge or falling edge?

Q starts at 0 . Follow the signals left to right. Which events can update Q?

Find the update events

At A, B, C, D, E and F, say whether Q changes or holds. Explain using the clock and D input.

A holds; B changes

A: D changes to 1, but CLK remains 0, so Q stays 0. B: CLK rises from 0 to 1, so Q copies D = 1 and changes to 1.

Both hold Q = 1

C: the clock falls (1 → 0), which does not trigger this flip-flop. D: D falls to 0 without a rising clock edge. Q still stores 1.

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

You Do · Work independently

BLACK PEN: Write your own answers. Do not look at the Review & Improve pages yet.

Question 1

TRACE · 4 MARKS

A rising-edge-triggered D-type flip-flop starts with Q = 0. Trace the different timing diagram and complete Q after each rising edge. [4]

Question 2

EXPLAIN · 3 MARKS

Explain the roles of the D input, clock input and Q output of a D-type flip-flop. [3]

Question 3

APPLICATION · 3 MARKS

A register must store an 8-bit value while combinational logic elsewhere in the CPU changes. Explain how D-type flip-flops can provide this storage. [3]

Question 4

ANALYSE · 4 MARKS

Q initially stores 1. D falls to 0 while CLK is LOW. The clock rises, and D returns to 1 while CLK stays HIGH. State Q after each event and explain why. [4]

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

Question 5

COMPARE · 3 MARKS

Explain two differences between an AND gate and a D-type flip-flop. [3]

Question 6

CHALLENGE · 4 MARKS

A 4-bit register initially stores 1010. Just before the next rising edge its D inputs are 0111. Explain the value stored before and after that edge, and what happens if the D inputs subsequently change without another rising edge. [4]

Exit Ticket · Answer before checking

If D changes but no clock trigger occurs, Q should keep its previously stored value.

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

Review & Improve · GREEN PEN

Now change to GREEN PEN. Find each original question. Tick correct answers; read the model and rewrite mistakes.

Page 1 onwards · Do Now

Q1. Sum = 1 because 1 XOR 0 = 1. Carry = 0 because 1 AND 0 = 0.

Q2. ¬A ∧ ¬B. Negate both inputs and change OR to AND.

Q3. 30 is the right child of 20: 30 is less than 40, but greater than 20.

Q4. The exponents already match. $0.1100_2 - 0.0100_2 = 0.1000_2$. The result is $0.1000_2 \times 2^3 = 4_{10}$.

You Do pages · Independent questions

Q1. Clocked state
CODE:
Q after triggers:
1, 1, 0, 1
Check: You change Q only at the listed triggers.
Working: at each rising clock edge, copy D to Q. Between clock edges, Q holds the previously stored value.

Q2. Purpose and inputs
CODE:
A D-type flip-flop stores one bit.
D is the data value to be stored.
CLK controls when D is sampled.
Q is the stored output and holds its value between triggers.
Check: Your answer separates the data input from the timing input.

Q3. APPLICATION · 3 MARKS
CODE:
A register made from eight D-type flip-flops stores eight bits, one bit per flip-flop. Each D input supplies its next bit; a shared rising clock edge copies the eight input bits to Q together. The Q outputs hold the byte between edges.
Check: Award marks for technically accurate statements tied to the scenario.

Q4. ANALYSE · 4 MARKS
CODE:
Initially Q = 1. (1) D changes to 0 while CLK is low: Q stays 1. (2) CLK rises with D =

0: Q changes to 0. (3) D changes to 1 while CLK remains high: Q stays 0. Q updates only on the rising clock edge.
Check: Award marks for technically accurate statements tied to the scenario.

Q5. COMPARE · 3 MARKS
CODE:
An AND gate is combinational: its output depends immediately on its current inputs and it stores no state. A D-type flip-flop stores one bit and Q only updates to D on the active clock edge; Q holds its value between edges.
Check: Award marks for technically accurate statements tied to the scenario.

Q6. CHALLENGE · 4 MARKS
CODE:
Before the rising edge, the register continues to store 1010 even when D becomes 0111. At the next rising edge, all four flip-flops copy their D inputs and the register stores 0111. Any further D changes without another rising edge leave the stored value at 0111.
Check: Award marks for technically accurate statements tied to the scenario.

Exit Ticket · Model answer

True. A D-type flip-flop holds its stored Q value until its triggering clock event.

Try the question again

Choose one question you got wrong. All correct? Choose the hardest.

1. Copy the whole original question:

2. Write a complete new answer in GREEN PEN. Show working if needed: